F70LB Life Insurance Maths:
Unit-linked polices

Heriot-Watt University, Edinburgh

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8 Unit-linked policies

8.1 Before we begin: Some housekeeping

Welcome to this chapter of the F70LB course in which we apply our emerging cashflow methods from the previous to chapter to a new type of product called unit-linked policies.

As before, parts of the notes that are also in the slides are in blue font. This is to help you follow the notes in class.

Please bring these notes with you to class so that we can talk through the exercises.

In this chapter, we do not use Excel examples. However, an essential skill you must develop is to model unit-linked policies in Excel. You will learn this in the tutorial sheets.

Let's begin ...

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8.2 Introduction

A traditional life insurance product is simple but not transparent. The policyholder sees only premiums and a sum assured, and perhaps bonuses, but not the underlying investment in real assets, the expenses and the surpluses.

A unit-linked (UL) policy presents the policyholder with explicit charges, of various kinds, from which the insurer must meet its expenses and make a profit; and directs the premiums less charges into investment funds of the policyholder's choice (e.g. UK equities, US equities, bonds, property, cash) whose gains and losses the policyholder shares explicitly. Thus UL business is transparent but complicated.

The rise of UL business driven by a desire for greater transparency, but the necessary complexity meant that it did not become feasible until computing power enabled individual investment accounts to be maintained, and enabled profit-testing of the policies. Unit-linked policies have been common for a long time now and have largely replaced traditional endowment assurances.

In some other countries, the term *equity-linked insurance* is used for the equivalent of UL.

8.2.1 Overview of unit-linked policies

Fundamental idea: It's a lot like a unit trust.

- Policyholders **buy "units"** of investments
  - . . . and typically have a **wide range of investment choice**, eg shares, bonds, overseas, domestic, etc.
- Each fund is split into **many identical units**
  - . . . and policyholders can all have different numbers of units
  - . . . which they **can buy and sell**
- The policyholder's units are normally held in a **separate fund**, segregated from the insurer's own assets. (This is called the "unit fund")
- It is mainly a **savings** product. The policy might also provide:
  - A guaranteed minimum death benefit, eg death benefit is max(sum assured, unit fund) for some fixed sum assured
  - A guaranteed minimum maturity benefit
- There are lots of **different types of charges** (and, as always, expenses). . .

8.2.2 Example of the basic idea

UL policies (also unit trusts, called mutual funds in the USA) work as in the following simple example.

In this example we ignore all charges and we assume there are no guarantees and no mortality.

On 1 January 2024, HW Equity Unit Fund has £70m of assets. The fund is split into many identical units. Policyholders can all hold different numbers of units. They can buy and sell units.

The unit price for that fund at that date is £2.00.

Questions to work through in class:

Q1(a): How many units are there in the fund?

Ms Jones decides to invest £4,000. The insurer accepts this investment and creates new units.

Q1(b): What is the new total value of the Unit Fund on 1 January 2024, immediately after this new policyholder invests their £4000?

Q1(c): . . . and how many units are there now in the Fund? How many units does the new policyholder have?

1 January 2026, unit fund investments have returned 25% since 1 January 2024, so the unit price has risen by the same amount to £2.50. Suppose the size of unit fund has fallen to £50m.

Q1(d): (Discuss) How can the unit fund be smaller even though the unit price has gone up?

Q1(e): How many units are now in the fund?

Suppose Ms Jones decides to sell half her units.

Q1(f): How much money does Ms Jones receive for selling half her units?

We will cover the solution in class and it will be included in the slides. If you like, you can write the solution in this blank space before proceeding to the following page.

8.2.3 The basic idea continued: Guarantees

Sometimes UL policyholders can choose to have a guaranteed minimum death benefit. The most common form is the *maximum* of the bid value of the unit fund (denote this F_t) and a guaranteed minimum sum assured (denote this S_t; it is often a constant S).

$$\begin{aligned}\text{Death benefit} &= \max(S_t, F_t) \\ &= F_t + \max(0, S_t - F_t)\end{aligned}$$

The insurer might also offer a guaranteed minimum investment return, e.g. 0% (i.e. no fall in the unit price.)

In our example, the policyholder's entire premium was invested in units and they got back all of those units at the end. **So how does the insurer make any profit?** The **insurer applies various charges**, which we discuss later. The charges need to cover the insurer's **expenses, profit and the cost of any guarantees**.

8.2.4 Understanding UL: Risk and profit

Now we have seen the basic idea of how UL works, let's check our understanding by thinking about risk and profit for these policies. Consider these questions:

Question 2: Who bears the investment risk?

a) The policyholder

b) The insurer

c) They share it equally

d) Both, but differently

Question 3: What (other) risks does the insurer have in respect of unit-linked policies?

Question 4: The insurer holds a separate fund of its own i.r.o. unit-linked policies, called the "non-unit fund" or "sterling fund". What might this need to cover?

We will cover the solution in class and it will be included in the slides. If you like, you can write the solution in this blank space before proceeding to the following page.

8.2.5 Visualising the unit fund and non-unit fund

We need to consider two funds. The *unit fund* tracks the policyholders' units. The *non-unit fund* (or *sterling fund*) tracks the insurers' cashflows i.r.o. the UL policies.

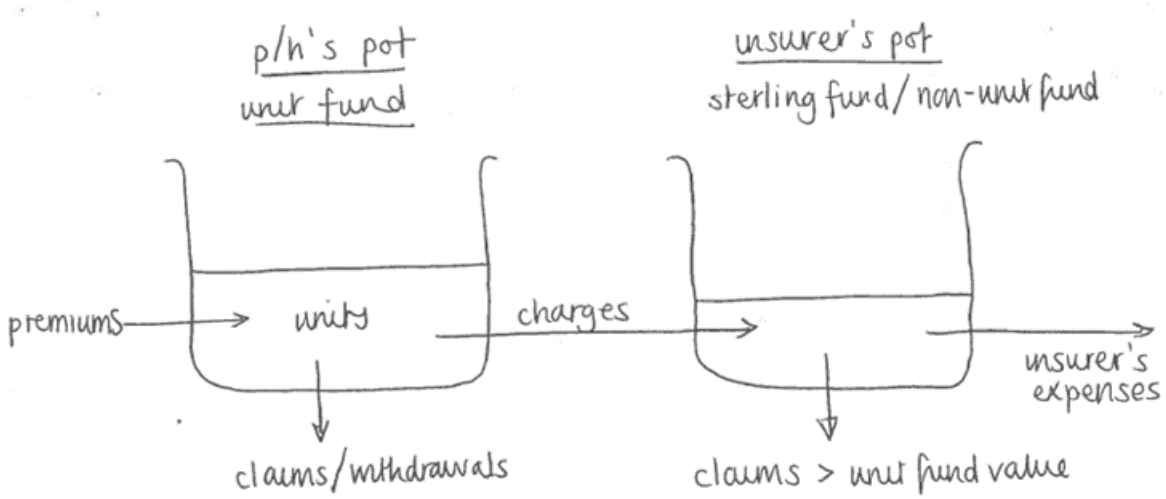


Figure 1: Unit and non-unit funds

The policyholder's premiums are paid into the unit fund where they buy units. Various charges are deducted from the unit fund and paid into the insurer's non-unit fund.

On maturity, death or surrender of the policy, the value of the policyholder's units are paid to the policyholder from the unit fund.

If there is a guaranteed minimum death benefit then there will be a deduction from the non-unit fund used to top-up the value of the units to the minimum level. (This is indicated by "claims > unit fund value" in the diagram.)

The main outflow from the non-unit fund is the insurer's expenses. For example, commission, underwriting costs, administration, IT, salaries and buildings.

8.3 Charges on UL policies

The insurer covers expenses and makes a profit by levying charges. Charges may be deducted from premiums before they are invested in units; or they may be deducted from the policyholder's fund of units.

To the extent that the insurer can *match* the charges deducted to the actual expenses incurred, they reduce the risk of loss.

We now consider different forms of charges.

8.3.1 Bid/Offer Spread

The policyholder cannot buy and sell units at the same price. Instead there are separate bid and offer prices:

- Offer price: Price to buy one unit
- Bid price: Price to sell one unit

The *bid/offer spread* λ is:

$$\lambda = 1 - \frac{\text{Bid Price}}{\text{Offer Price}}$$

For example, we could have $\lambda = 0.05$ or 5%. The effect of the bid/offer spread is that a proportion λ of each premium is taken as a charge as it is paid.

8.3.2 Allocation rate

Separately from the bid-offer spread, only a defined proportion a_t of the premium P_t payable at time t is used to buy units, at the offer price. Therefore $a_t\ P_t$ is invested and, allowing for the bid-offer spread, the actual charge levied is $(1 - a_t)P_t + \lambda\ a_t\ P_t$. The proportion a_t is called the *allocation rate or factor*.

- Early years: a_t is low, perhaps even 0%.
- Later years: a_t is high, perhaps over 100%.

8.3.3 Policy fee

The policy fee is a monetary amount deducted from the policyholder's fund on a regular basis (e.g. monthly or annually). For example, £25 per annum.

8.3.4 Fund management charge (FMC)

The Fund Management Charge (FMC) is a regular percentage deduction from the policyholder's unit fund . For example, 1% a year.

In example calculations, we usually assume the FMC is deducted from the unit fund at the end of each year, just before the payment of any benefits.

8.3.5 Mortality charge

We noted above that the death benefit often has this form:

$$\begin{aligned}\text{Death benefit} &= \max(S_t, F_t) \\ &= F_t + \max(0, S_t - F_t)\end{aligned}$$

The mortality charge (MC) pays the expected cost of the additional benefit above the unit fund, also called the *Expected Death Strain*, for the preceding month or year.

$$\begin{aligned} MC &= q_{x+t-1}(S_t - F_t) && \text{if} \quad (S_t > F_t) \\ MC &= 0 && \text{if} \quad (S_t \leq F_t) \end{aligned}$$

In practice, the MC is usually deducted every month, but for simplicity, we assume it is deducted at the end of the year, after the FMC is levied.

Some UL policies with a death benefit have no MC, but have a lower allocation rate instead.

8.3.6 Surrender penalty

Recall that there is often new business strain early in a policy and profits only appear later. The charges in the early years of a policy are often too small to cover the high inital expenses, although profits are made in the later years.

Hence, the insurer will often apply a surrender charge if a policyholder leaves early. This is to recover the shortfall to date, and/or to recover the future profits it would have made. It may be a percentage of the unit fund which depends on duration, or it could be a fixed monetary amount.

Example:

YR	Surrender Penalty % of Fund
1	50%
2	30%
3	10%
≥ 4	0%

8.3.7 UL charges: Test your understanding

To wrap on charges, let's do a couple of quizzes.

Quiz: For each charge, suggest some matching expenses.

1. Bid/offer spread
2. Allocation
3. Policy fee
4. Fund management charge
5. Mortality charge
6. Surrender penalty

We will cover the solution in class and it will be included in the slides. If you like, you can write the solution in this blank space before proceeding to the following page.

Here is our second quiz on charges:

1. **Question:** Suppose we set the allocation rate (and other charges) to match expected expenses as closely as possible. What might be the problem with this?
2. **Question:** The aim of profit-testing with conventional policies is to set the premium. What do we aim to set by profit-testing unit-linked policies?
3. **Question:** What's the most natural way to determine the reserve in the unit fund? (Non-unit fund is harder.)

We will cover the solution in class and it will be included in the slides. If you like, you can write the solution in this blank space before proceeding to the following page.

8.3.8 Further comments on charging structures

We conclude our section on charges with some additional explanation.

Charging structures come in many forms and it is necessary to ***read the question carefully*** to make sure cashflows are calculated ***in the right order***. For example, policy fees can be levied at the start or end of the year, as can FMCs. If two charges are applied at the same time (e.g. at year end), the question will tell you which to calculate first; likewise whether deaths and surrenders are dealt with before or after charges.

As a rule, the closer the charges match the expenses they are meant to cover, the better. This reduces risk to the office — whenever outgo precedes income there is risk to the office. Hence:

- Initial commission and expenses $\Rightarrow$ low allocation factor in year 1.
- Regular servicing expenses $\Rightarrow$ policy fee.
- Fund management expenses $\Rightarrow$ FMC.

However some of these charges can make the policy look unattractive, especially low allocation factors or *front-end loading*. Hence offices may change the charging structure, e.g. by having level allocation factors but higher policy fees and FMCs. Such changes usually result in profits emerging later, which reduces profitability and increases risk.

8.4 Profit-testing UL policies

The objective in pricing conventional business is to set the **premium** rates to meet our profit criteria under the premium basis. The objective in pricing unit-linked business is to set the **charging structure** to meet our profit criteria under the premium basis. You will again practice these calculations using Excel in tutorials, and you will be assessed on Excel calculations. However, to give you practice learning the ideas we will now work through parts of an example by hand.

8.4.1 The unit fund calculation

We start by projecting the unit fund. Consider the following unit-linked policy.

- Five-year term
- Premium: £5,000 a year, annually in advance
- Survival Benefit: Bid value of units at the end of the term
- Death Benefit: maximum of £20,000 and the bid value of units, paid at the end of the year of death, after deduction of FMC and MC.

The charges are:

- Allocation Rate: 70% in year one and 102% each year after that
- Bid/Offer Spread: 5%
- Policy Fee: £30 deducted from units at the start of each year
- FMC: 1% of the bid value of units, taken at the end of each year
- Death Charge: 1% of the sum at risk at the end of year, after the FMC

The pricing basis is:

- Unit growth rate: 8% a year
- Mortality: $q_x = 0.01$ at all ages.

Use the following column headings and see how far you can get with a calculator filling in the unit fund cashflows:

Yr	(A)Alloc'd Premium	(B) Policy Fee	(C) Year-End Fund	(D) FMC	(E) MC	(F) Year-End Fund (2)
1 Etc						

HINTS: Every entry should be a £ amount. (A) We want the premium that the policyholder actually gets added to their "account", ie you need to allow for allocation rate and bid-offer spread (B) Just the £ amount each year (C) Don't forget investment return and last year's year-end fund. What else? (D) Use your answer in column (C) with FMC info from the question (E) This is the hardest one. Calculate expected value of this year's "sum at risk", ie q_{x+t-1} times (death benefit minus unit fund) (F) Use (C), (D), (E) and take care to get the signs right!

We will cover the solution in class and it will be included in the slides. If you like, you can write the solution in this blank space before proceeding to the following page.

8.4.2 The non-unit fund calculation

If we know the unit fund, and the charges each year, we can work out the non-unit fund cashflows. We need more assumptions in the pricing basis:

- Expenses: assume 40% of the first premium, and in each year thereafter, £20 plus 2.5% of each year's premium
- The rate of interest earned on sterling cashflows: assume 4% a year. Note that the sterling fund is the *insurer's* assets and may be invested cautiously to reduce risk.
- Mortality: 0.5% per annum.

Then the cashflows to and from the sterling fund in year t, denoted SCF_t, are:
Charges − Expenses + Interest − Benefits
where 'benefits' means death benefits in excess of the unit fund. These correspond to the elements of the profit vector in a profit test.

Use the following column headings and see how far you can get with a calculator filling in the non-unit fund cashflows:

Yr	Unalloc Premium	Expense	Pol Fee	Int	FMC	Mort Charge	Mort Cost	SFC_t
1		2000.00	30.00		35.59	164.77		
2		145.00	30.00		88.27	112.61		
3		145.00	30.00		145.17	56.29		
4		145.00	30.00		206.60	0.00		
5		145.00	30.00		272.90	0.00		

8.4.3 The profit signature

Once we have SFC_t we can find the profit signature:

$$\sigma_t = {}_{t-1}p_x \times SFC_t$$

and evaluate the profitability of the policy in the usual way using EPV of profit, IRR, DPP, profit margin and any other criteria we are interested in.

8.5 Reserves for UL polices

The insurer needs to hold reserves in respect of the unit fund and the non-unit fund.

Unit reserve

We have already discussed the unit fund, and noted that the unit fund reserve should normally just be the matching type of assets with the same value as the unit fund.

As long as the unit funds each hold the right amount of assets of the right type, the office runs no risk, no matter how badly the assets in the fund perform. This is because the changes in asset values are reflected precisely in the unit price, which is passed on to the policyholder. Hence the appropriate reserve is just the matching assets.

If the office deviated from this strategy, e.g. by holding assets of a different kind from those advertised; or a different quantum of assets; it runs a mismatching risk. For example, if the office held US equities in a UK equity fund, and the UK stock market rise while that in the USA fell, it might be obliged to make good the difference to its customers, who quite reasonably would expect the value of their units to depend on the UK stock market.

Sterling or Non-Unit Reserve The starting point for reserving for non-unit cashflows is a profit signature σ'_t, where the dash indicates that it is calculated using the valuation actuary's basis (which must satisfy the regulator).

The valuation basis need not be the same as the pricing basis (and when carrying out a valuation many years after the policy was sold, we should not expect it to be) but σ'_t will be calculated in exactly the same way.

The question then is: are any elements σ'_t of the valuation profit signature negative? If so, then there is an anticipated need to apply capital to the support of the policy in future, and prudence demands that that need must be reserved for now.

Elements σ'_t that are positive do not worry us and need not be reserved for, indeed doing so would lead to negative reserves which we know to avoid.

We allow for future negative cashflows by setting up a reserve now, that will meet the future negative cashflows as they arise. For example, suppose the valuation interest rate is 4% per annum, and the valuation profit signature is:

$\sigma = (-100, -50, 35, 65, 150)$

(which will automatically allow for the mortality of the valuation basis).

Then the reserve we ought to set up at time 0 is:

$100(1.04)^{-1} + 50(1.04)^{-2}$

which will meet the negative items as they arise. This process is often called *zeroisation of negative cashflows*.

8.6 Unitised with-profits

We will now briefly introduce unitised with-profits.

Revision exercise: What are the main features of conventional (i.e. non-unitised) with-profits policies?

We will cover the solution in class and it will be included in the slides. If you like, you can write the solution in this blank space before proceeding to the following page.

Unitised with-profits (UWP) policies are like non-profit unit-linked policies, except that the change in the total value of the policyholder's units is determined by a bonus, not directly by the underlying investments.

Like conventional with-profits, the bonus is discretionary.

There are two methods to add the bonus to the value of the units:

- Increase the **price** of the units in line with the bonus
- Increase the **number** of the units in line with the bonus

UWP has similar charges to unit-linked policies, so (unlike conventional with profits) the size of the bonus is mainly determined by (smoothed) investment performance.

8.7 Summing up

Key skills

On completion of this chapter of the course, you should be able to:

- Explain UL and UWP policies and their main features
- Undertake profit-tests for UL and UWP policies, including projecting the unit fund and non-unit fund
- Evaluate different charging structures and link them to expenses
- Calculate UL reserves by zeroising negative cashflows

You also need to be able to implement these calculations in Excel. This is covered in tutorials.

What next?

This is the end of our notes for the first half of F70LB. Make sure that you complete all the tutorial exercises, so that you can implement these ideas in Excel. Ensure that you revise the ideas used in this section of the course, and that you have a solid understanding of the material covered last semester, too, so that you are prepared for your F70LB assessment. Good luck!